ROADWATCH · OXHACK

Dashboard Demo Instructions

Presenter 2 · timed actions, verified screen states and recovery steps

Your role. Presenter 1 speaks the whole time. You operate the laptop silently: you advance the slides and drive the live dashboard in sync with the speech. Never speak over Presenter 1. If something fails, follow the recovery column and keep going.

What the demo shows. One **retrospective** forecast week, Monday **16 March 2026**, using the default inspection list (strict single-block LightGBM, top-K). This is a replay of saved historical predictions, not a live maintenance recommendation.

Every value below was verified. The steps were replayed headlessly against the running app (`scripts/final/verify_run_of_show.py` → `outputs/final/verification/run_of_show_values.json`). Screenshots come from the real dashboard.

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1. Before the session (do this at least 10 minutes before)

1.1 Start the dashboard

```
cd C:\Users\drish\Downloads\Roadwatch
.venv\Scripts\python.exe -m streamlit run app.py --server.headless true --server.port 8501
```

Leave this window open: closing it stops the dashboard. Startup takes a few seconds. Check it is up:

```
(Invoke-WebRequest -UseBasicParsing http://localhost:8501/_stcore/health).Content     # expect: ok
```

1.2 Open the demo week in the browser

1. Open **Microsoft Edge** at `http://localhost:8501/?week=2026-03-16&view=map`.
2. Press **F11** for full screen. Keep the browser zoom at 100% (**Ctrl+0**).
3. Confirm the opening state (Table A).

1.3 Warm up every step once

The first time a view or capacity loads it takes a few seconds; later loads are instant. Do the whole demo once, then reset:

1. Click **2% (469 blocks)**, then **1% (235 blocks)**.
2. Scroll to **Why each block is listed**.
3. Click **Decision-policy comparison**, then **Inspection worklist & map**.
4. **Reset:** reload `http://localhost:8501/?week=2026-03-16&view=map` and scroll so the map and the worklist table fill the screen. The four metrics should be just visible at the top.

1.4 Slides

1. Open `outputs/final/RoadWatch_OxHack_Pitch.pptx` in PowerPoint and start the slideshow from Slide 1 (**F5**).
2. Practise switching once: **Alt+Tab** to the browser, **Alt+Tab** back to the slideshow, then → or **Space** for the next slide.
3. **Backup:** open `outputs/final/RoadWatch_OxHack_Pitch.pdf` in a second Edge tab, so slides can be shown from the browser if PowerPoint misbehaves.

Table A: opening state (week 16 Mar 2026, default settings)

Where	What you should see
Sidebar · Forecast week (Monday)	**Mon 16 Mar 2026**
Sidebar · Inspection capacity	**1% (235 blocks)** selected
Sidebar · Inspection list	**Strict single-block LightGBM, top-K (best under conservative expected capture)** selected
Metrics row	Forecast week starting **16 Mar 2026** · Blocks to inspect **235** · Capacity scenario **1%** · Priority P1 / P2 / P3 **47 / 71 / 117**
Caption under metrics	"List: Strict single-block LightGBM, top-K … | selection rule: **top-K**."
Model version caption	primary__LGBM | retrain 2026-03-02 | model sha256 1e8448d92f01 …
Map	blue thick lines = 235 selected blocks; red/orange lines = rest of the top 20% of the ranking
Map caption	"Map shows 235 selected blocks (blue, thick) and 4,690 background blocks."
Worklist, first rows	1 · P1 · 17 AVE · 11204 · 0.266 — 2 · P1 · FLATLANDS AVE · 11236 · 0.250 — 3 · P1 · 39 ST · 11232 · 0.204

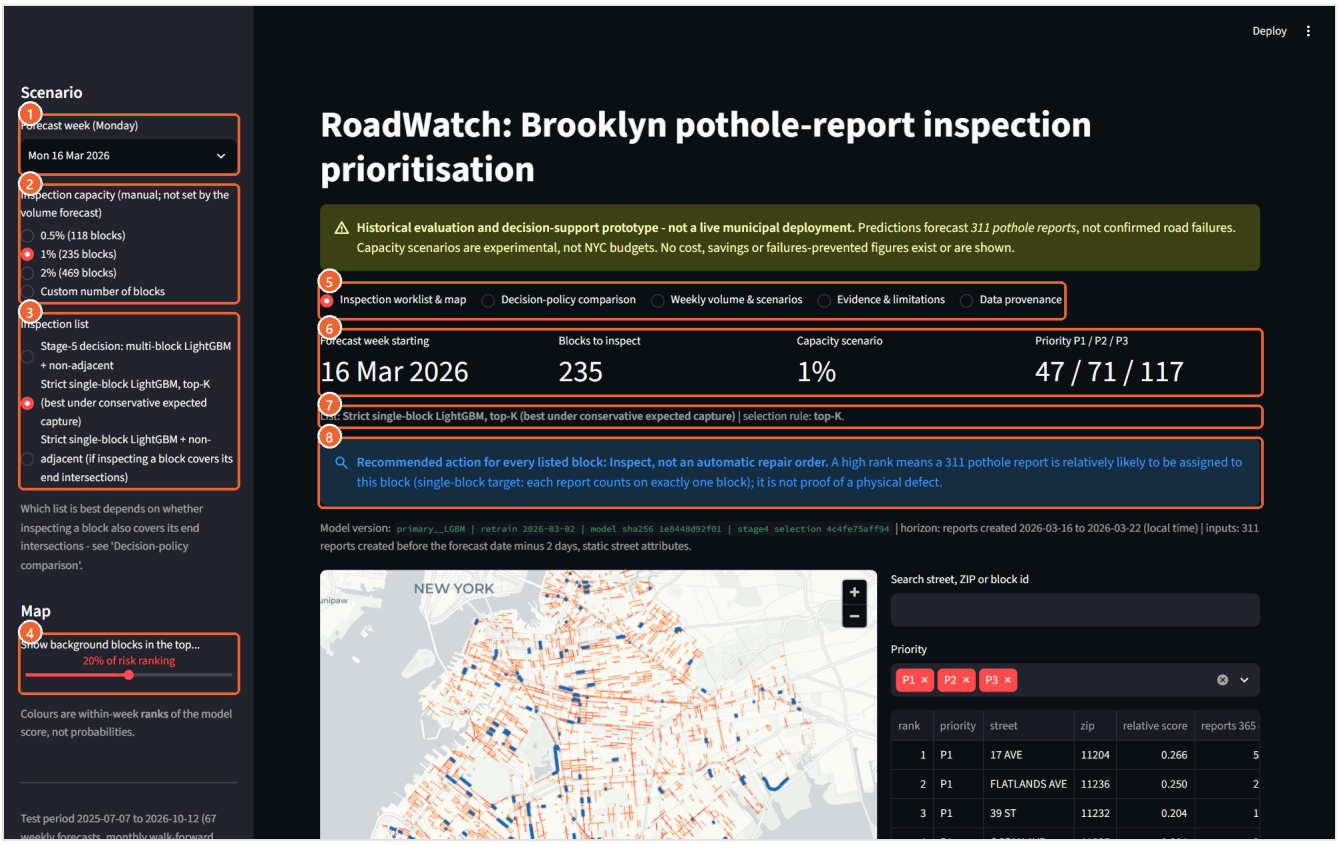


Figure 1. Opening state. Controls: (1) forecast week, (2) inspection capacity, (3) inspection list, (5) view selector; (6) the metrics row confirms the state.

2. Timed actions

All times are elapsed time from Presenter 1's first word. Wait for each **cue phrase** before acting; the clock is only a guide.

Time	Cue phrase (Presenter 1)	Your action	Audience sees	If it goes wrong
0:00	(starts speaking)	Slide 1 already on screen	Title slide	—
0:25	"…goes where it is needed?"	→ Slide 2	Data collection	—
0:55	"…typically four."	→ Slide 3	Processing	—
1:27	"…weather from five days before."	→ Slide 4	Prediction	—
1:57	"…reported before."	→ Slide 5	Evidence	—
2:27	"…exactly one block."	→ Slide 6	Decision-making	—
2:40	"Let's see it in the dashboard."	**Alt+Tab** to the browser (already on the map view, week 16 Mar 2026)	Map, worklist, metrics **235 / 1%**	Browser not responding → stay on Slide 6 (see §4)
2:42	"…ranked in the table."	Hover over the top of the worklist table (rank 1, 17 AVE)	Ranked table	—
2:52	"If the budget doubles to 2 percent…"	Sidebar → click **2% (469 blocks)**	Blocks to inspect **469**; P1/P2/P3 **94 / 141 / 234**; more blue lines	Spinner over 3 s → keep going; Presenter 1's line still holds
2:58	"…469 blocks."	Sidebar → click **1% (235 blocks)**	back to **235**	—
3:00	"Every block shows its reasons…"	Scroll down to **Why each block is listed** (block **#1 17 AVE** is preselected)	"#1 17 AVE (ZIP 11204, block 52878) – P1" and its rationale	If another block is shown, open the **Block** dropdown and pick **#1 17 AVE**
3:15	"One subtlety: what counts as catching…"	Scroll to the top, click the view **Decision-policy comparison**, then scroll so the yellow **Audit finding** box and the table below it are visible	Four-definition table (Test, 1%)	View slow to load → leave it; Presenter 1 continues
3:25	"…spreading inspections across intersections looks best."	Point the cursor at **Strict single-block LightGBM + non-adjacent → Intersection-node coverage 17.3%**	—	—
3:32	"Without that assumption the strict list wins…"	Point at **Strict single-block LightGBM (top-K) → Fractional (expected) 13.5%**	—	—
3:43	"…do not treat it as confirmed." (or "…our default." if the optional line is skipped)	**Alt+Tab** to the slideshow and press →	Slide 7 · Limitations	Slideshow lost → open the PDF tab, go to page 7
4:08	"…no failures prevented."	→ Slide 8	What it would take to deploy	—
4:30	"…Thank you."	Do nothing. Leave Slide 8 up for questions	—	—

What the four-definition table shows (Test, 67 weeks, 1% capacity)

Read the **Fractional (expected)** and **Intersection-node coverage** columns:

List (row label in the dashboard)	Fractional (expected)	Intersection-node coverage
Strict single-block LightGBM (top-K) — **default**	**13.5% (12.5-14.5)**	16.4% (15.3-17.5)
Strict single-block LightGBM + non-adjacent	12.3% (11.5-13.2)	**17.3% (16.3-18.4)**
Stage-5 decision: multi-block LightGBM + non-adjacent	11.3% (10.6-12.1)	**17.3% (16.3-18.4)**
Historical-rate baseline	8.4% (7.8-9.1)	10.8% (10.0-11.6)

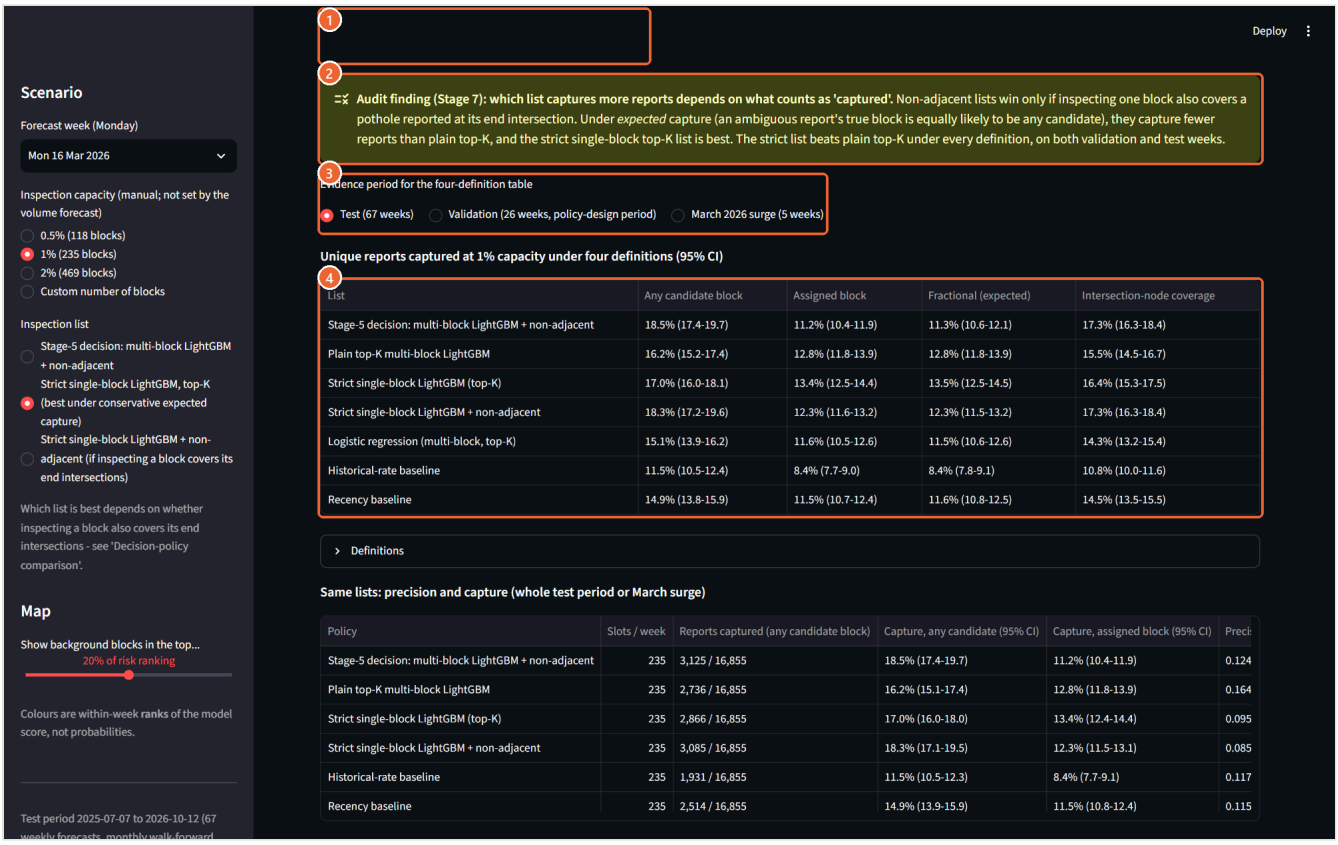


Figure 2. Decision-policy comparison. (2) the audit-finding box; (4) the four-definition table you point at.

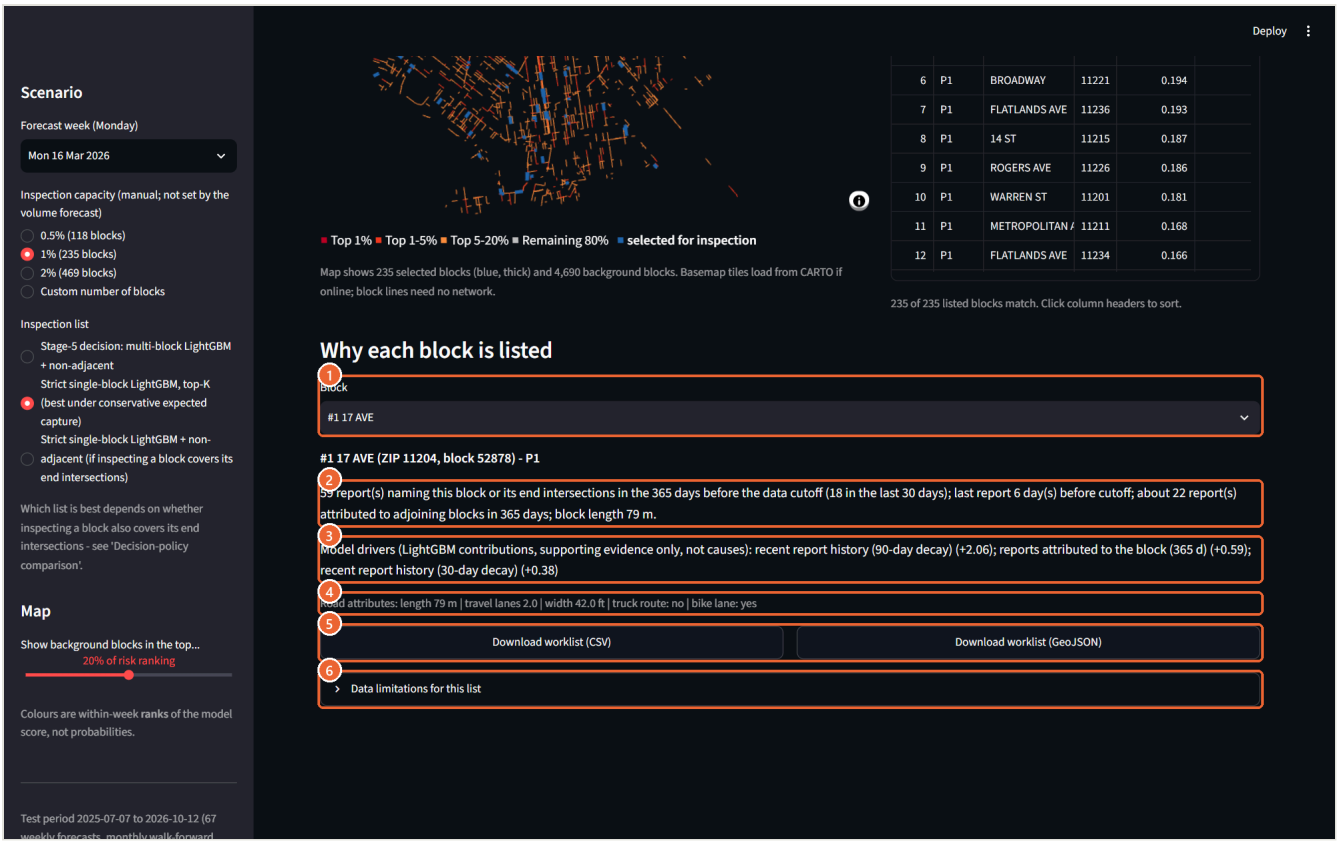


Figure 3. "Why each block is listed" for block #1 (17 AVE): rationale (2) and model drivers (3).

3. Extra steps if time permits (for example during Q&A)

Goal	Action	Verified result
Show the real test results	Click **Evidence & limitations**	Metric pairs (any candidate / fractional): **Strict single-block top-K at 1%: 17.0% / 13.5%**; **Stage-5 decision at 1%: 18.5% / 11.3%**
Show the March 2026 stress case	Click **Weekly volume & scenarios** (week still 16 Mar 2026)	Red box: volume model did not beat simple baselines; MAE table row **March 2026: 355.9** (last-7-days **295.2**), 80% interval coverage **40%**, mean actual **843**; metric "Context for 2026-03-16: forecast reports (80% interval)" **593 (411-806)**
March capture by policy	View **Decision-policy comparison** → *Evidence period for the four-definition table* → **March 2026 surge (5 weeks)**	Strict top-K **16.9%** any candidate / **12.9%** fractional; historical rate **13.1%** / **9.2%**
Search the worklist	On the map view, type **FLATBUSH** in *Search street, ZIP or block id*	"9 of 235 listed blocks match."
Export the worklist	On the map view, scroll to the bottom → **Download worklist (CSV)** and **Download worklist (GeoJSON)**	Files `roadwatch_worklist_2026-03-16_strict_topk_1pct.csv` (235 rows) and `.geojson` (235 features), saved to the browser's Downloads folder
Compare lists on the map	Sidebar → *Inspection list* → **Stage-5 decision: multi-block LightGBM + non-adjacent**	Caption changes to "selection rule: **top-K + non-adjacent (validation-decided)**"; switch back afterwards

4. Recovery and fallbacks

Problem	What to do
Map tiles (street basemap) do not load (no internet, slow Wi-Fi)	Carry on: nothing else is affected. The blue and red block lines, the worklist, the charts and the downloads are all local; only the grey street background comes from the internet (CARTO). Verified with all external hosts blocked. Presenter 1 does not need to mention it.
A view shows a spinner for more than 5 seconds	Do not click repeatedly. Wait for Presenter 1's next cue; if it is still loading, move on to the next step.
Wrong week or settings after a click	Reload `http://localhost:8501/?week=2026-03-16&view=map` (**Ctrl+L**, paste, **Enter**). That resets to Table A in about 3 seconds.
"A required saved output is missing ..." error	Do not fix it live. Switch to the slides (fallback below). After the talk, run the command shown in the message.
Dashboard will not start or the browser crashes	Stay on **Slide 6**: it shows a genuine dashboard screenshot of the same week plus the coverage-assumption chart. Presenter 1 uses the fallback lines in the run-of-show. At 3:43 advance to Slide 7 as normal. If asked, Appendix A1 (Slide 9) has the full four-definition table.
PowerPoint freezes	Switch to the Edge tab with `RoadWatch_OxHack_Pitch.pdf` and use **Page Down**.
Port 8501 already in use	Start with `--server.port 8502` and open `http://localhost:8502/?week=2026-03-16&view=map`

Before the session, check the full demo still behaves as documented:

```
.venv\Scripts\python.exe -I -X utf8 scripts\final\verify_run_of_show.py      # expect: problems: []
```